RP KARONGI college **le 29 jan ,2026**

BACKEND USING JAVA

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Project description

Project Title: Online Document Hub

1. Executive Summary

The Online Document Hub is a web-based application developed to support efficient, secure, and centralized document management for organizations such as schools, companies, cooperatives, and clinics. The system allows users to upload, access, edit, download, and share documents through the internet using a standard web browser. By digitizing document management processes, the system reduces paperwork, improves accessibility, enhances collaboration, and strengthens data security through controlled user access. The solution is designed to be reliable, scalable, and suitable for academic and organizational environments.

2. Project Objectives

The main objective of this project is to design and implement a secure Online Document Hub that enables efficient document management over the internet. The specific objectives of the project are to allow users to upload, view, edit, download, and share documents online, to implement role-based access control to enhance security, to provide centralized document storage, to support collaboration among authorized users, and to ensure data integrity, availability, and backup.

3. Project Scope

The scope of the Online Document Hub focuses on the development of a web-based system that can be accessed through internet browsers. It includes user authentication and authorization, document upload, storage, retrieval, editing, and sharing, as well as administrative management of users and access rights. The project does not include offline document access or native mobile application development, which may be considered in future enhancements.

4.Literature Review

The Online Document Hub is a web-based application designed to enable efficient management, access, and sharing of documents for organizations such as schools, companies, cooperatives, and clinics. Literature indicates that web-based Document Management Systems (DMS) significantly enhance accessibility, collaboration, and security of organizational information.

Alade (2023) highlights the design and implementation of web-based DMS, focusing on automated document handling, improved productivity, and centralized access. Triyana and Fianty (2023) demonstrate that web-based DMS improves efficiency in educational institutions by enabling users to upload, view, edit, and share documents through browsers. Bresfelean et al. (2017) emphasize that DMS adoption reduces operational costs and enhances information governance.

Collaboration and real-time access are essential in document management. Wong et al. (2012) and Tuuk et al. (2024) report that web-based DMS enhance productivity by supporting version control, workflow management, and document editing. Features such as real-time editing and update tracking promote teamwork in distributed settings.

Security and access control are critical for safeguarding sensitive organizational data. Ozman (2025) reviews secure DMS architectures, highlighting the need for robust access control and encryption. Sultan et al. (2020) demonstrate that Role-Based Access Control (RBAC) models provide fine-grained permissions, ensuring only authorized users can access, modify, or delete documents. Integrating these security measures strengthens system reliability.

Cloud-based DMS improve scalability, reliability, and remote access. IJERT (n.d.) discusses cloud integration in DMS, highlighting centralized storage, disaster recovery, and web accessibility without additional software. Vaandering (2003) emphasizes web-based architecture's role in ensuring cross-platform access, aligning with the goals of the Online Document Hub.

Existing platforms such as Mayan EDMS, OpenKM, and LogicalDOC validate the feasibility of web-based document hubs. They offer features like centralized storage, versioning, workflow management, and secure access, supporting the design choices of the Online Document Hub.

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5. Innovation

The Online Document Hub introduces an innovative approach by offering centralized document management specifically designed for institutions. It provides customizable role-based access control, secure browser-based access without requiring additional software installation, and a scalable architecture that can grow with organizational demands.

6. Methodology

The project follows the System Development Life Cycle methodology. The development process begins with requirements analysis to understand user needs, followed by system design to define architecture and components. Implementation involves coding and database integration, after which the system undergoes testing and validation. Finally, the system is deployed and maintained to ensure continued performance and reliability.

7. Actors

The system involves three main actors. The administrator is responsible for managing users, assigning roles, controlling permissions, and maintaining the system. Registered users can upload, view, edit, download, and share documents based on their assigned roles. In some cases, a guest user may be allowed to view shared documents with limited access.

8. User Story Table

User stories describe how different actors interact with the system. Administrators need the ability to manage users and roles in order to control system access. Users require the ability to upload documents so they can store information securely online. Users also need to download

documents for later use and share documents to collaborate with others. Administrators must be able to monitor system activity to maintain security and accountability.

9. Expected Outcomes

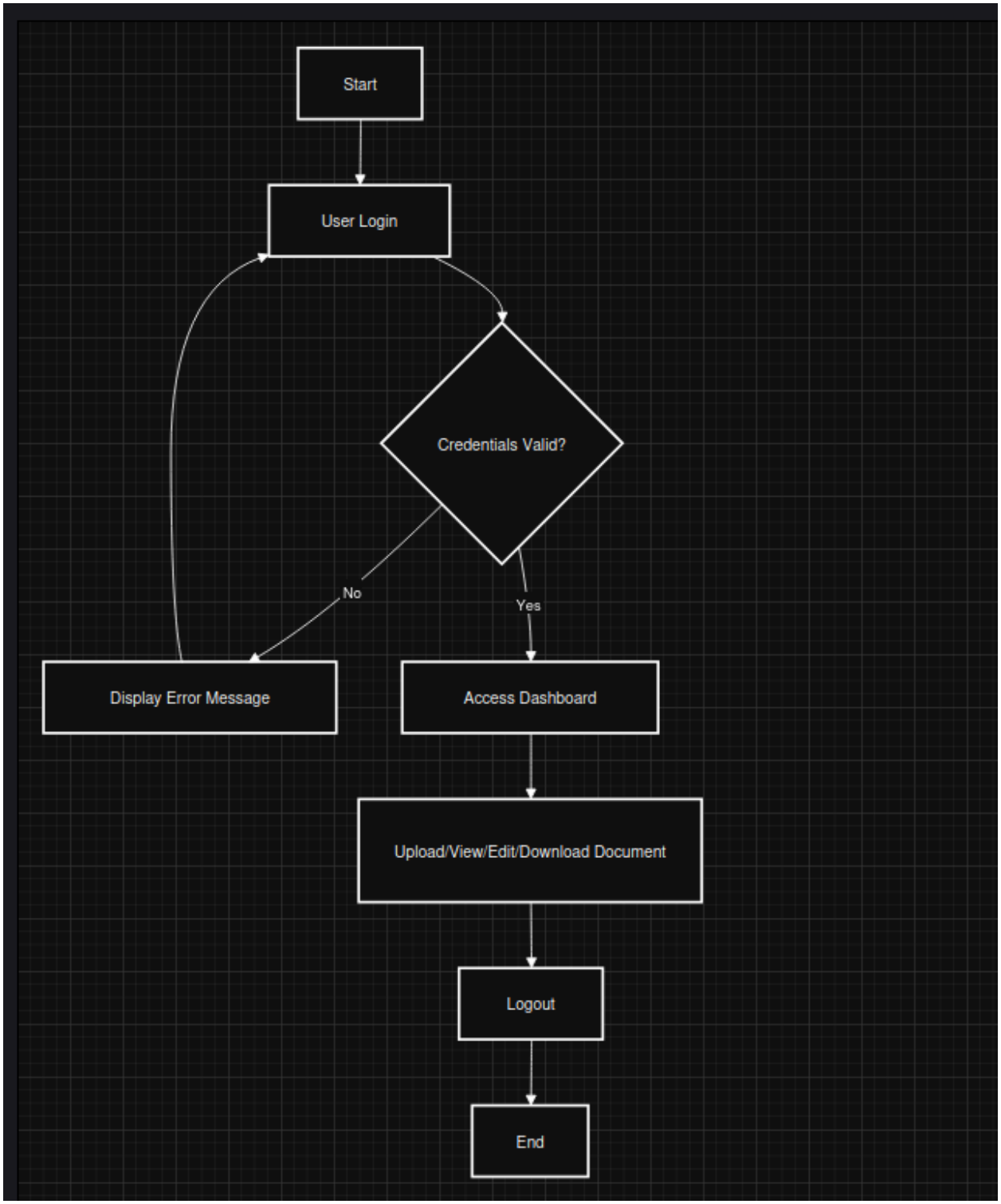
The expected outcome of this project is a fully functional Online Document Hub that improves document accessibility, security, and management efficiency. The system will reduce document duplication and loss while enhancing collaboration and productivity within organizations.

10. Flowchart

The following flowchart illustrates the overall system workflow from user login to logout.

flowchart TD

https://drive.google.com/file/d/1rjp0sfhUQj5nNFrHZCwoFGsze3A599wt/view?usp=sharing



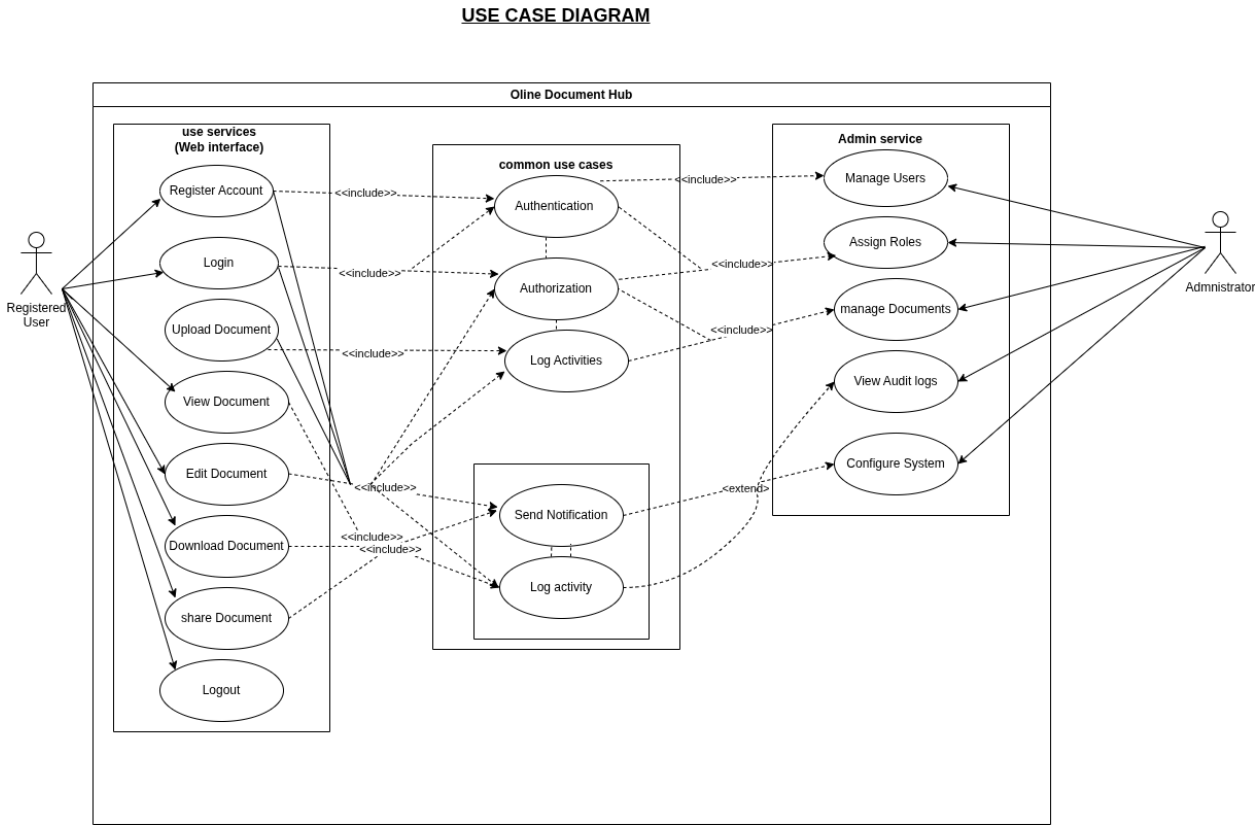
The system then authenticates the user and verifies their role. Once authenticated, the user is directed to the dashboard where document-related operations such as uploading, viewing,

editing, downloading, and sharing can be performed. The process ends when the user logs out of the system.

11. Use Case Diagram

The following use case diagram represents interactions between system actors and the Online Document Hub.

https://drive.google.com/file/d/1mokDMIqAVKVQ2SFWXTS8BuZD9_e5VfAA/view?usp=sharing



12. Entity Relationship Diagram

The Entity Relationship Diagram below shows the relationship between the main system entities.

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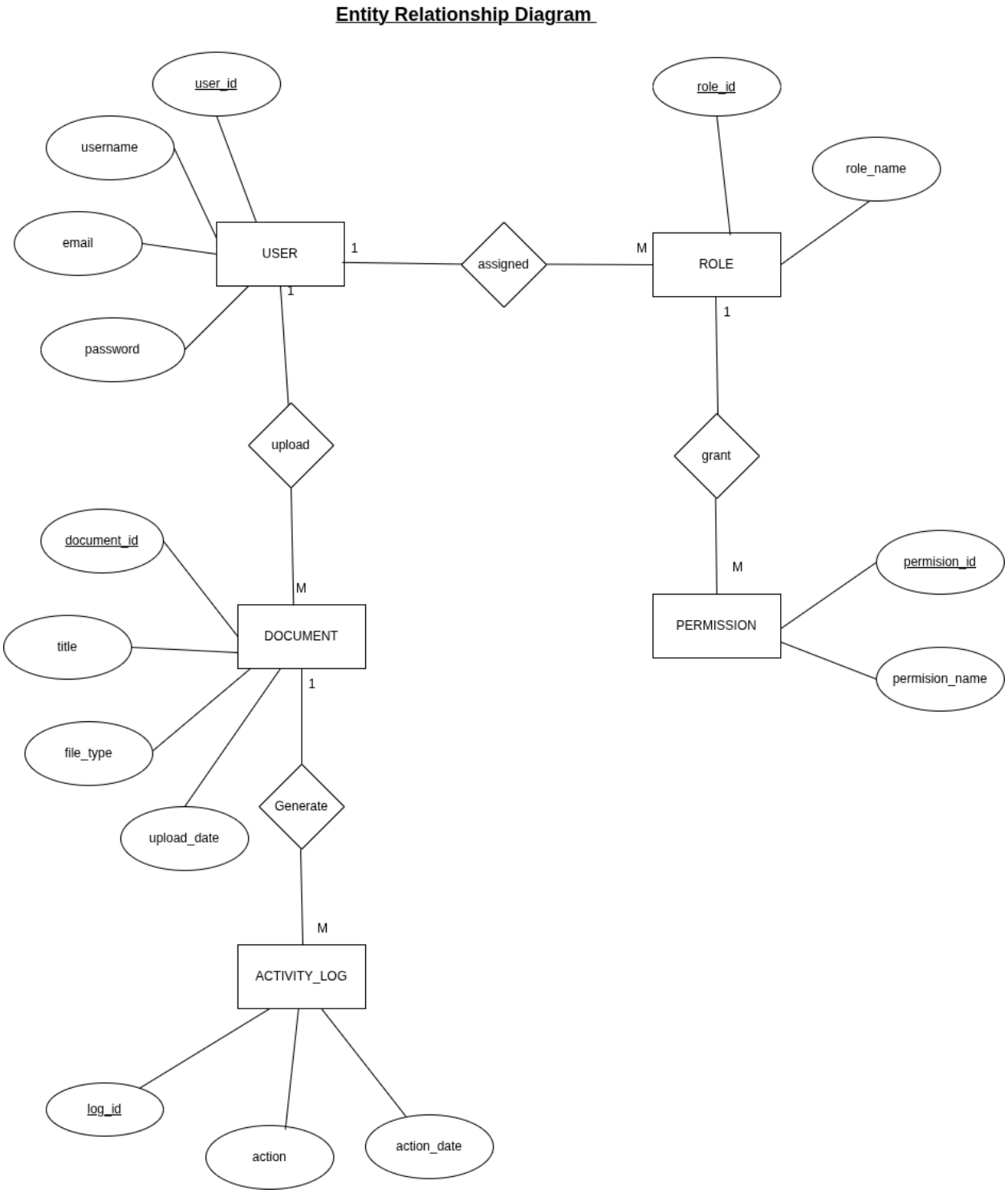

13. System Architecture Diagram

The following diagram illustrates the three-tier system architecture of the Online Document Hub.

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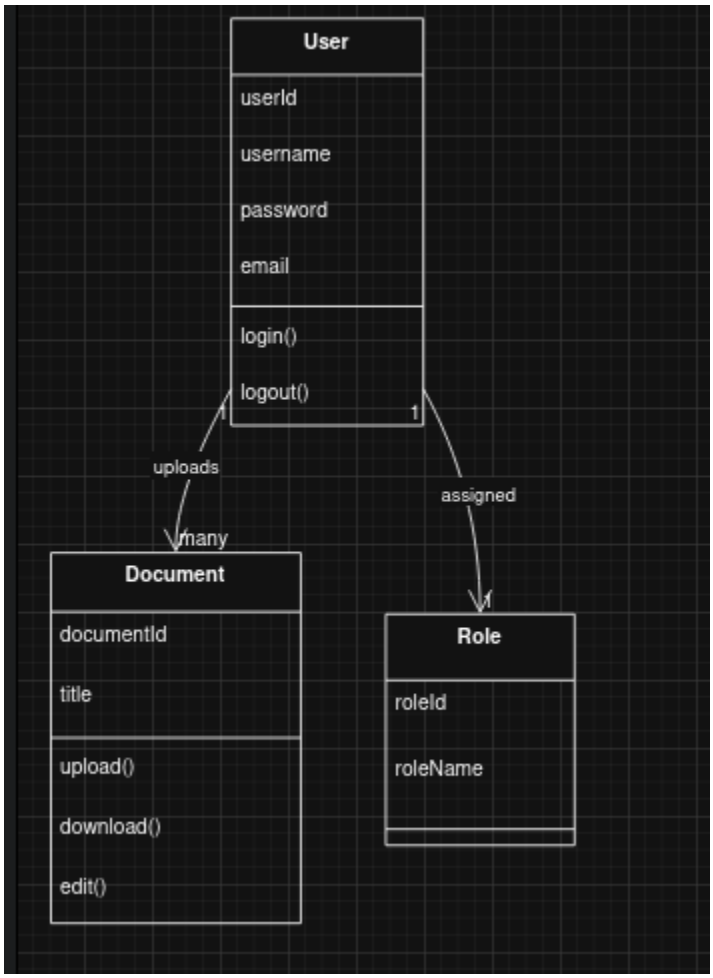
14. Class Diagram

The class diagram below represents the main system classes and their relationships.

https://drive.google.com/file/d/1uv1MYw9tjHtBBvSI3GYUiE4bNVLlhWt3/view?usp=sharing



15. Functional Requirements

FR-1: The system shall provide a secure user registration mechanism to allow authorized individuals to create user accounts.

FR-2: The system shall authenticate users using valid credentials before granting access to any system functionality.

FR-3: The system shall allow authenticated users to upload documents to a centralized online repository.

FR-4: The system shall enable users to view documents based on their assigned roles and permissions.

FR-5: The system shall allow authorized users to edit documents while maintaining data integrity.

FR-6: The system shall permit authorized users to download documents from the system.

FR-7: The system shall support document sharing among users according to defined access rights.

FR-8: The system shall implement role-based access control to regulate user permissions and actions.

FR-9: The system shall allow administrators to create, update, assign roles to, and delete user accounts.

FR-10: The system shall enable administrators to manage documents and monitor system activities for security and accountability purposes.

16. Non-Functional Requirements

NFR-1: The system shall ensure data security through robust authentication and authorization mechanisms.

NFR-2: The system shall protect stored documents and user information from unauthorized access.

NFR-3: The system shall be reliable and operate with minimal downtime.

NFR-4: The system shall be available to users through the internet whenever required.

NFR-5: The system shall be scalable to accommodate future increases in users and document volume.

NFR-6: The system shall provide acceptable performance with efficient response times for all user operations.

NFR-7: The system shall present a user-friendly and intuitive interface suitable for users with basic computer skills.

NFR-8: The system shall be compatible with commonly used web browsers.

User Roles

The Online Document Hub System defines user roles in order to control access to system functionalities and ensure secure and efficient document management. Based on the use case diagram, the system supports three main roles: Guest User, Registered User, and Administrator. Each role has clearly defined permissions that align with the system's functional requirements.

1. Guest User

A Guest User is an unauthenticated user who has not yet logged into the system. This role represents new or temporary users accessing the platform for the first time.

According to the use case diagram, a Guest User is allowed to register an account and log in to the system through the web interface. Guest Users do not have access to document management functionalities such as uploading, viewing, editing, downloading, or sharing documents. They also do not have access to administrative services or system configuration features.

2. Registered User

A Registered User is an authenticated user who has successfully logged into the system and accesses the system through the web interface.

Based on the use case diagram, a Registered User can upload documents, view documents, edit documents, download documents, and share documents. The user can also log out of the system

after completing their activities. All actions performed by a Registered User are subject to authentication, authorization, and activity logging as shown in the common use cases of the diagram. Registered Users do not have permission to manage users, assign roles, view audit logs, or configure the system.

3. Administrator

The Administrator is a privileged user responsible for managing users, documents, and system settings.

As shown in the use case diagram, the Administrator can manage users, assign roles, and manage documents within the system. The Administrator also has permission to view audit logs and configure system settings. Administrative actions are protected by authentication and authorization mechanisms and are logged for security and accountability purposes.

NO	**User Role**	**Description**	**Authorized Use Cases / Responsibilities**
1	Registered User	An authenticated user who uses the system through the web interface to manage documents.	Register account, Login, Upload document, View document, Edit document, Download document, Share document, Logout

2	Administrator	A privileged user responsible for managing users, documents, and system configuration.	Manage users, Assign roles, Manage documents, View audit logs, Configure system
3	System (Online Document Hub)	The system itself that handles internal operations automatically.	Authentication, Authorization, Log activities, Send notifications
4	Authentication Service	A logical system component that verifies user identity.	Authenticate user credentials
5	Authorization Service	A logical system component that checks user permissions.	Authorize user actions
6	Logging Service	A background system service for security and auditing.	Log activities, Store audit logs
7	Notification Service	A system service responsible for	Send notifications

		sending alerts to users.	
8	Web Interface	The user-facing component through which users interact with the system.	Provide access to user services
9	Admin Service Module	A system module that supports administrative operations.	Support user and document management
10	Audit Module	A system component used to monitor and review system activity.	View audit logs

17. Expected Exceptional Cases and Proposed Solutions

Possible exceptional cases include invalid login credentials, network interruptions, unauthorized access attempts, and file upload failures. These issues will be addressed by displaying appropriate error messages, retrying failed operations, denying unauthorized access while logging events, and notifying users when file uploads fail.

18. Project Timeline

The project is planned to be completed over a period of approximately seven weeks. One week will be allocated for requirements analysis, one week for system design, three weeks for implementation, one week for testing, and one week for deployment.

19. Project Team and Assigned Tasks

The project team consists of a project manager responsible for planning and coordination, a developer responsible for system implementation, a database administrator responsible for database design and maintenance, and a tester responsible for system testing and validation.

20. Future Work

Future enhancements to the Online Document Hub may include the development of a mobile application, integration with external cloud services, advanced document search and analytics features, and the use of artificial intelligence for document classification.

21. Conclusion

The Online Document Hub provides a secure and efficient solution for managing documents over the internet. By centralizing document storage and enforcing controlled access, the system improves organizational efficiency, collaboration, and data security. The project meets academic requirements and provides a solid foundation for future expansion and improvement.